WORKED SOLUTION

STEP I 1988, Question 13

Mechanics • Elastic Strings and Collisions

STEP I 1988, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q13

The key idea

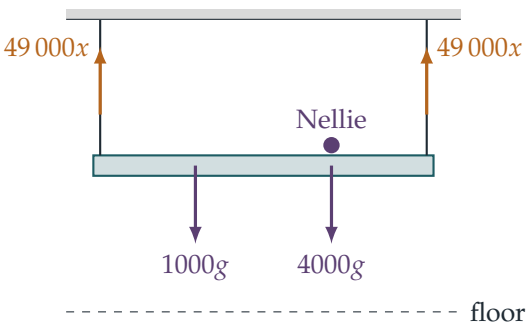
There are three stages, and each one has its own conservation law. Nellie's landing is a sudden inelastic collision, so momentum is conserved but energy is not. The descent to the floor involves only gravity and the elastic ropes, so energy is conserved. After Nellie steps off, energy is conserved again, and we have to ask whether the stored elastic energy is enough to lift the plank past the point where the ropes are at their natural length.

Solution

We model the plank as a rigid body that stays horizontal, so we can treat it as a particle of mass 1000 kg moving vertically. Nellie is a particle of mass 4000 kg, and the ropes are light elastic strings obeying Hooke's law. Both ropes always have the same extension x, so together they exert an upward force

$$2 \cdot \frac{490\,000}{10}\, x = 98\,000\, x \text{ N}.$$

So the pair of ropes acts like a single spring of stiffness $K = 98\,000 \text{ N m}^{-1}$, with elastic energy $\frac{1}{2}Kx^2 = 49\,000\, x^2$ J.



The starting position

Before Nellie arrives the plank hangs in equilibrium, so the extension e_0 satisfies

$$98\,000\,e_0 = 1000 \times 9.8 \quad \Longrightarrow \quad e_0 = 0.1 \text{ m}.$$

The landing

Nellie lands and stays on the plank, so this is a perfectly inelastic collision. The impact takes a very short time, and the ropes' tension and the weights are finite forces, so their impulse during the impact is negligible. Momentum is therefore conserved, and the common speed v just afterwards satisfies

$$4000 \times 5 = 5000\,v \quad \Longrightarrow \quad v = 4 \text{ m s}^{-1}.$$

Ask yourself Can we use conservation of energy straight from Nellie's arrival?

No. When an elephant lands on a plank and stays there, a lot of kinetic energy turns into heat and sound: here $\frac{1}{2} \cdot 4000 \cdot 5^2 = 50\,000\,\text{J}$ becomes $\frac{1}{2} \cdot 5000 \cdot 4^2 = 40\,000\,\text{J}$, so $10\,000\,\text{J}$ is lost. Energy is conserved only from the instant just after the landing.

The descent

Let h be the plank's initial height above the floor. Plank and Nellie (total mass $5000\,\text{kg}$) start with speed $4\,\text{m s}^{-1}$ and come to rest at the floor, a distance h lower, where the extension is $0.1 + h$. The only forces doing work are gravity and the ropes, so

$$\underbrace{\tfrac{1}{2} \cdot 5000 \cdot 4^2}_{\text{KE lost}} + \underbrace{5000 \cdot 9.8\,h}_{\text{GPE lost}} = \underbrace{49\,000[(0.1+h)^2 - 0.1^2]}_{\text{elastic energy gained}},$$

$$40\,000 + 49\,000\,h = 49\,000\,(0.2h + h^2).$$

Dividing by $49\,000$ and rearranging,

$$h^2 - 0.8\,h - \tfrac{40}{49} = 0, \qquad \text{so} \qquad h = 0.4 + \sqrt{0.16 + \tfrac{40}{49}} = \frac{14 + 2\sqrt{299}}{35},$$

taking the positive root because $h > 0$. Hence

$$h \approx 1.39 \text{ m}.$$

The ropes stay taut throughout this stage, because the extension only grows as the plank goes down.

After Nellie alights

Nellie steps off gently, so the plank alone ($1000\,\text{kg}$) is at rest at the floor with extension

$$X = 0.1 + h \approx 1.488 \text{ m}.$$

The ropes go slack if the plank rises far enough for the extension to fall to zero, which means rising a height X. While the ropes are taut, energy is conserved, and so the plank reaches that level if the stored elastic energy exceeds the gain in potential energy:

$$\tfrac{1}{2}KX^2 = 49\,000\,X^2 \approx 108\,500\text{ J}, \qquad 1000 \cdot 9.8 \cdot X \approx 14\,600\text{ J}.$$

The first is far bigger, so the plank arrives at the natural-length position still moving upwards, with kinetic energy of about 93 900 J (a speed of about $13.7\,\mathrm{m\,s^{-1}}$). Above this point the ropes would have to be shorter than their natural length, so they are slack. **Yes, the ropes become slack.**

Ask yourself Couldn't we just use simple harmonic motion?

Yes, while the ropes are taut. The plank alone performs SHM about its equilibrium extension 0.1 m, and it starts $h \approx 1.39$ m below it, so the amplitude is 1.39 m. The top of the oscillation would be at extension $0.1 - 1.39 < 0$, which is impossible for a taut rope. So the ropes must go slack before the plank reaches the top. (In fact the SHM argument gives the answer as soon as $h > 0.1$.)

Once the ropes are slack the plank moves freely under gravity, rising a further $13.7^2/(2 \times 9.8) \approx 9.6$ m. This is less than 10 m, so (just!) the plank doesn't reach the ropes' fixing points before falling back.

Checks

- Substituting $h = 1.3881$ into the energy equation: both sides equal 108 017 J, to the nearest joule.
- Stepping $\ddot{x} = g - Kx/5000$ numerically from $x = 0.1$, $\dot{x} = 4$ gives the plank stopping after descending 1.3881 m.
- Stepping the plank's motion after Nellie leaves shows the extension reaching zero while the plank is moving up at $13.7\,\mathrm{m\,s^{-1}}$, and a highest point 11.07 m above the floor, which matches $X + 9.58$.
- Units: 490 000 N / 10 m gives $\mathrm{N\,m^{-1}}$, so every term in the energy equation is in joules.

Common slips

- Using energy conservation through the landing, which gives $\sqrt{20} \approx 4.47\,\mathrm{m\,s^{-1}}$ for the combined speed instead of $4\,\mathrm{m\,s^{-1}}$.
- Measuring elastic energy from the natural length but forgetting the initial extension 0.1 m: the energy gained is $49\,000[(0.1 + h)^2 - 0.1^2]$, not $49\,000\,h^2$.
- Using the stiffness of one rope instead of two.

Beyond STEP

The pair of ropes is a nonlinear spring: it pulls when stretched but cannot push when compressed. Systems like this, which switch between different equations of motion, are called piecewise-linear or "impact" oscillators, and even simple ones can behave in surprisingly complicated ways. Kleppner and Kolenkow, *An Introduction to Mechanics*, has a careful treatment of energy, collisions and oscillations that underpins all of this question.

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